

Math 54 Discussion Worksheet

Name: _____

Date: _____

Problem 1

Let u and v be vectors in $\mathbb{R}^n$. Convince yourself that the parallelogram defined by u and v can be described as the set P of all linear combinations $au + bv$ where $0 \leq a, b \leq 1$. Then, let $T : \mathbb{R}^n \rightarrow \mathbb{R}^m$ be a linear transformation. Show that any point $p \in P$ maps to a point in the parallelogram defined by $T(u)$ and $T(v)$.

Problem 2

Let $T : \mathbb{R}^n \rightarrow \mathbb{R}^m$ be a linear transformation and let $\{v_1, v_2, v_3\}$ be a linearly dependent set. Prove that the set $\{T(v_1), T(v_2), T(v_3)\}$ is also linearly dependent.

Problem 3

Let $T : \mathbb{R}^n \rightarrow \mathbb{R}^m$ be a linear transformation. Show that T can be expressed as an $m \times n$ matrix.